Question Paper Preview

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Change Font Color :	No
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Deep Learning

Section Id :	640653124551
Number of Questions :	12
Section Marks :	40
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531859432 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : DEEP LEARNING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536024932. ✔ YES

6406536024933. ✖ NO

Question Number : 2 Question Id : 6406531859433 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
What is the primary advantage of word embeddings compared to one-hot encoding?
Options :
6406536024934. ✖ Simpler implementation
6406536024935. ✔ Captures semantic relationships
6406536024936. ✖ They automatically correct spelling errors
6406536024937. ✖ Word embeddings are sparser than one-hot encoding

Question Number : 3 Question Id : 6406531859434 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
What is the primary advantage of Transformer encoders over RNN encoders for long sequences?
Options :
6406536024938. ✖ Lower memory usage
6406536024939. ✔ Parallelization during training
6406536024940. ✖ Smaller model size
6406536024941. ✖ Simpler architecture

Question Number : 4 Question Id : 6406531859435 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Suppose that we implement an OR function with two boolean inputs using the McCulloch Pitts (MP) neuron. Assume that the neuron does not have any inhibitory input. Which of the following option(s) represents correct θ and the Number of Correctly Classified (NCC) data points for various values of threshold θ.

$$\hat{y} = \begin{cases} 1, & \text{if } \sum x_i \geq \theta \\ 0, & \text{otherwise.} \end{cases}$$

Options :

6406536024942. ✖ $\theta = 0$, NCC = 4

6406536024943. ✔ $\theta = 1$, NCC = 4

6406536024944. ✖ $\theta = 1$, NCC = 3

6406536024945. ✖ $\theta = 2$, NCC = 1

6406536024946. ✔ $\theta = 2$, NCC = 2

6406536024947. ✖ $\theta = 3$, NCC = 0

Question Number : 5 Question Id : 6406531859436 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following optimizers use momentum (or momentum-like mechanisms) in updating gradients?
Options :
6406536024948. ✔ NAG
6406536024949. ✖ AdaGrad
6406536024950. ✖ RMSProp
6406536024951. ✖ AdaDelta
6406536024952. ✔ ADAM

Question Number : 6 Question Id : 6406531859437 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are true for the saturated neurons problem? Select all that apply.

Options :

6406536024953. ✔ ReLU activation function is preferred over tanh, as it is less likely to saturate.

6406536024954. ✖ Tanh is preferred over sigmoid as it solves the problem of neuron saturation.

6406536024955. ✔ Leaky ReLU mitigates the problem of neuron saturation compared to ReLU.

6406536024956. ✔ If a neuron saturates in a network, the weights do not get updated further.

Question Number : 7 Question Id : 6406531859438 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

You are training a deep feedforward neural network (100 layers) for binary classification with a sigmoid output with tanh activation in some hidden layers and ReLU activations in some hidden layers. You notice that gradients in some layers vanish early, preventing those weights from updating, even though the network hasn't converged.
Which of the following fixes could help?

Options :

6406536024957. ✖ Increase the size of your training set

6406536024958. ✔ Replace ReLU activations with leaky ReLUs

6406536024959. ✔ Use appropriate weight initialization

Question Number : 8 Question Id : 6406531859439 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

A transformer with a single self-attention layer has batch size $B = 4$, context length (sequence length) $T = 12$, and number of heads $h = 2$. How many attention scores are computed in total across two heads and all sequences for one forward pass?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1152

Question Number : 9 Question Id : 6406531859440 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

In a time-series prediction task, an LSTM network is used to encode input sequences where each time step is represented by a 3-dimensional feature vector ($x_t \in \mathbb{R}^3$). The LSTM computes its internal gates and states at each time step using the following equations:

$$
\begin{aligned}
i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i) \\
f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) \\
o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) \\
\tilde{s}t &= \tanh(W_s x_t + U_s h_{t-1} + b_s) \\
s_t &= f_t \odot s_{t-1} + i_t \odot \tilde{s}_t \\
h_t &= o_t \odot \sigma(s_t) \\
\hat{y}_t &= V h_t + c
\end{aligned}
$$

Here, $\odot$ denotes element-wise multiplication. The hidden state has dimension $h_t \in \mathbb{R}^4$, and the output vector at each time step satisfies $\hat{y}_t \in \mathbb{R}^3$.

If the LSTM is applied to sequences of length 5 ($T = 5$), compute the total number of trainable parameters in the entire network (including all weight matrices and biases).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

143

Question Number : 10 Question Id : 6406531859441 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Given the input matrix X and kernel K:

$$X = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 2 & 1 & -1 \\ 1 & 0 & -1 & 2 \\ 2 & 1 & 0 & -2 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

• Perform convolution of K over X with stride $= 1$ and no padding to obtain matrix A.
• Apply average pooling over A of size 2×2 with stride $= 2$ and no padding to obtain B.
• Apply the **ReLU activation** on B to obtain the final output $\hat{y}$.
Given that
$\frac{\partial L}{\partial \hat{y}} = -3,$
compute
$\frac{\partial L}{\partial K_{11}},$
where K_{11} is the **center element** of the kernel.
Submit your answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
-1.53 to -1.47

Question Id : 6406531859442 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (11 to 15)

Question Label : Comprehension

You are designing a Skip-Gram model to learn word embeddings. In this iteration, the model focuses on a specific **center word** and attempts to predict the surrounding **context words** within a window size of $d = 3$.

The sentence is: **Deep neural networks learn complex data patterns.**

We are currently processing the center word: **networks**.

Vocabulary = {0: complex, 1: data, 2: deep, 3: learn, 4: models, 5: networks, 6: neural, 7: patterns }

The one-hot vector for the center word (x_{center}) is a column vector of size $V = 8$.

$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}^T$

The model's weight matrices are given below:

Input/Center Weight Matrix ($W_{in} \in \mathbb{R}^{V \times k}$):

$$W_{in} = \begin{bmatrix} 0.5 & 1 \\ 1 & 0.5 \\ -1 & -1 \\ 0 & 2 \\ 1.5 & -0.5 \\ \mathbf{1} & \mathbf{-2} \\ -0.5 & 1 \\ -1 & 1 \end{bmatrix}$$

(Note: Row 5 corresponds to word index 5 `networks')

Output/Context Weight Matrix ($W_{out} \in \mathbb{R}^{k \times V}$):

$$W_{out} = \begin{bmatrix} 1 & -1 & -2 & 2 & 0 & 1 & 1.5 & -1 \\ 2 & -1.5 & 1 & 3 & -2 & -1 & -0.5 & 2 \end{bmatrix}$$

For the center word w_c, the hidden layer representation is given by:

$h = W_{in}^T \cdot x_{center}$

(Or simply selecting the row corresponding to the center word).

The unnormalized score vector z for all words in the vocabulary is calculated as:

$z = W_{out}^T \cdot h$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 11 Question Id : 6406531859443 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

In the Skip-gram model with a window size of $d = 3$ (on each side), how many **unique** input and output pairs will be generated for the following sentence. The sentence is: **deep neural networks learn complex data patterns.**

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 12 Question Id : 6406531859444 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Calculate the hidden layer representation vector h for the center word "networks".

Options :

6406536024964. ✖ $\begin{bmatrix} 1.5 \\ -0.5 \end{bmatrix}$

6406536024965. ✔ $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

6406536024966. ✖ $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$

6406536024967. ✖ $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Question Number : 13 Question Id : 6406531859445 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

We want to predict the immediate **next word** to the word "networks" in the sentence, which is "learn". Calculate the unnormalized score (z_{learn}) for this specific target word.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
-4.00

Question Number : 14 Question Id : 6406531859446 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Identify the word in the vocabulary that has the highest probability of being predicted as a context word given the center word 'networks'. Provide the corresponding **integer index (key)** from the vocabulary list as your answer.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
4

Question Number : 15 Question Id : 6406531859447 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
How many total weights from W_{in} and W_{out} combined will be updated?
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
18

Question Id : 6406531859448 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (16 to 18)

Question Label : Comprehension

Suppose you are given three encoder hidden states at time t:

$$h_j = h_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad h_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad h_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

The previous decoder hidden state is:

$$s_{t-1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Given the attention score function:

$$\text{score}(s_{t-1}, h_j) = V_{att}^{\top} \tanh(U_{att} s_{t-1} + W_{att} h_j)$$

where the hyperbolic tangent function is defined as:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

and

$$V_{att} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad U_{att} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad W_{att} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 16 Question Id : 6406531859449 Question Type : SA

Correct Marks : 2

Question Label : Short Answer Question

Compute the attention score for hidden state h_1 using the given function. Submit the final answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.84 to 1.92

Question Number : 17 Question Id : 6406531859450 Question Type : SA
Correct Marks : 4
Question Label : Short Answer Question

Normalize the attention scores using the softmax function to obtain the attention weights α_{tj}. Submit α_{t1} (i.e., first element of the α vector). Submit the final answer correct to two decimal places.

$$\alpha_{tj} = \text{align}(s_{t-1}, h_j) = \frac{\exp\left(\text{score}(s_{t-1}, h_j)\right)}{\sum_{i=1}^{n} \exp\left(\text{score}(s_{t-1}, h_i)\right)}$$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
0.39 to 0.47

Question Number : 18 Question Id : 6406531859451 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

Calculate the context vector c_t:

$$c_t = \sum_{j=1}^{n} \alpha_{tj} h_j$$

Provide the sum of all the elements of c_t. Submit the final answer correct to two decimal places.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.68 to 1.76